

A Cost-Efficient Multi-Threshold Gas Leakage Detector Using an MQ-2 Sensor and LM339 Comparators

Abstract—The presence of liquefied petroleum gas (LPG), methane, hydrogen and volatile organic vapors as gaseous substances continue to pose threats of fire, explosion, and health complications. Though the commercial detectors are good, their high price and installation needs are a limitation to their application in less resource endowed areas. The paper describes an inexpensive, proof of concept gas leakage detector based on an MQ-2 metal-oxide semiconductor (MOS) gas sensor and an LM339 quad comparator that can be set to issue multiple alerts on varying thresholds. The system provides visual and audible cues, with tri-color LEDs and a buzzer, respectively, to alert to safety, caution and danger levels. Field adjustments of sensitivity can be made by the use of references based on potentiometers. Procedures A prototype was created on a breadboard and the total bill of materials cost around 564 BDT. Qualitative tests indicate the correct changes of state in various levels of exposure, and the design can be further extended without difficulties to enclosures and IoT telemetry. This type of system targets at residential, laboratory, automotive and light industrial safety applications where both cost and flexibility are important.

Index Terms—Gas leakage detector, MQ-2, LM339, comparator thresholds, low-cost sensing, safety systems.

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I. INTRODUCTION

Combustible gas leakage in domestic and industrial environmental health is a serious safety issue. Early detection and alarming helps avoiding the development of fires or explosion. mitigate health hazards. However, up-front device cost and In the case of complexity it can limit the deployment density, especially in low-resource contexts. This paper fills that gap by. a skeleton, bare-boned architecture that will incorporate an MQ-2 To realize, a 2 MOS sensor with an LM339 quad comparator. discrete, interpretable thresholds and low-latency alarming. Contributions: (i) a cost-transparent detector architecture with multi-threshold indication; (ii) a tunable comparator network permitting environment sensitivity; (iii) qualitative. validation and cost comparison indicating viability compared to commercial units; and (iv) an implementation pathway for future calibration and IoT-enabled alerts.

Contributions: (i) a cost-transparent detector architecture with multi-threshold indication; (ii) a tunable comparator network enabling environment-specific sensitivity; (iii) qualitative validation and cost comparison indicating viability relative to commercial units; and (iv) an implementation pathway for future calibration and IoT-enabled alerts.

TABLE I
BILL OF MATERIALS AND ESTIMATED COST

Component	Qty	Cost (BDT)
MQ-2 Gas Sensor	1	210
LM339 Quad Comparator	1	5
10k Ω Trimmer	3	20
220 Ω Resistor	3	5
470k Ω Resistor	1	10
Buzzer (5 V)	1	5
5 V Power Supply	1	50
LEDs (R, Y, G)	3	3
Breadboard	2	150
Wires/Misc.	–	110
Total	–	564

II. RELATED WORK

Little cost combustible gas monitoring often takes advantage of. MOS sensors (e.g., MQ-series) due to their broad analyte sensitivity and affordability. Most recent publications have shown IoT-enabled LPG house and industry monitoring [3], survey papers on gas and fire detection integration [4], and ESP32 or Arduino embedded applications. for real-time alarming [5], [6]. Datasheets pro-manufacturer. vide operating characteristics and conditioning recommendation networks; the present design aligns with these practices while focusing on reproducibility and tunable thresholds modularly. [1], [2]

III. MATERIALS AND METHODS

A. Bill of Materials

Table I Table I lists the fundamental elements. The MQ-2 output voltage, which is proportional to gas concentration, becomes conditioned by. the LM339 comparators. Reference levels are set. by a resistor divider with trimmer potentiometers for in-situ tuning. Unambiguous is given by LED indicators and a buzzer. operator feedback..

B. Development Cycle

Table II summarizes the staged development process.

IV. SYSTEM ARCHITECTURE AND OPERATING PRINCIPLE

Fig. 1 presents the block diagram of the proposed system, while Fig. 2 depicts the detailed schematic. The MQ-2 sensor outputs an analog signal proportional to detected gas concentration. This is fed into LM339 comparators, each configured with different reference voltages set by trimmers to establish

TABLE II
DEVELOPMENT CYCLE OVERVIEW

Phase	Activities
1	Literature review; MQ-2 and LM339 study
2	System design; schematic capture
3	Component procurement; cost verification
4	Breadboard implementation and integration
5	Qualitative exposure tests; sensitivity tuning
6	Documentation; preparation for submission

Block Diagram of Proposed Gas Leakage Detector

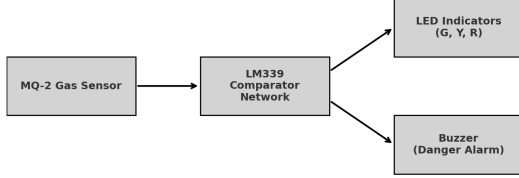


Fig. 1. Block diagram of the proposed gas leakage detector showing signal flow from sensing to alarm indicators.

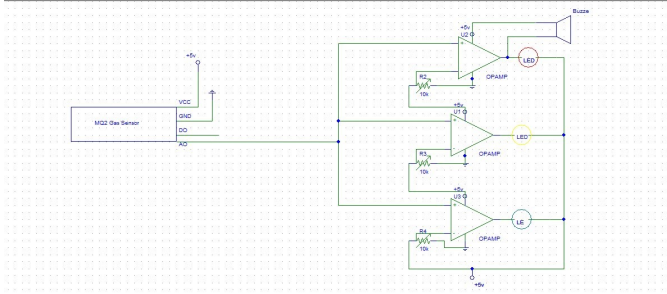


Fig. 2. Schematic of the detector using an MQ-2 sensor and LM339 comparators with multi-threshold outputs.

safe, caution, and danger thresholds. The comparator outputs drive green, yellow, and red LEDs respectively, with the danger stage also activating a buzzer.

V. IMPLEMENTATION

A proof-of-concept prototype was implemented on a breadboard. Fig. 3 shows the physical setup. Thresholds were adjusted conservatively so that in clean air only the safe indicator was active. Under moderate exposure the caution indicator asserted, and at strong exposure the danger indicator and buzzer activated.

VI. EVALUATION AND DISCUSSION

A. Qualitative Validation

The system showed desired state transitions. While quantitative calibration to reference gas standards was not performed, qualitative response established functional correctness of the comparator-based architecture.

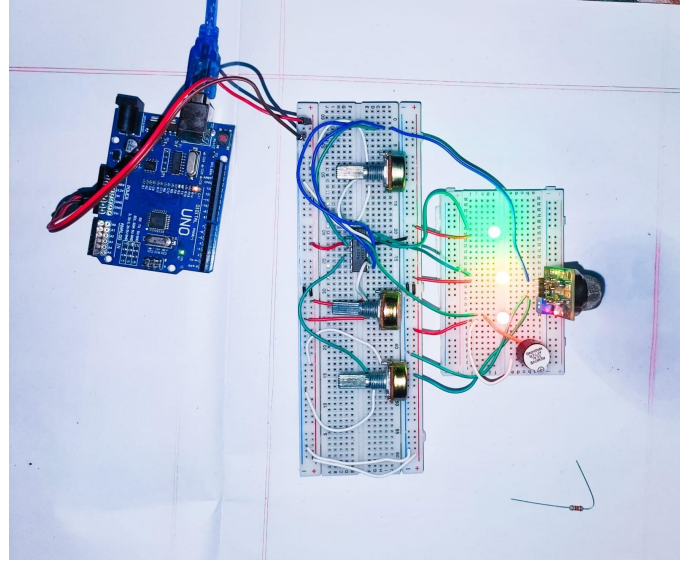


Fig. 3. Prototype implementation on a breadboard, illustrating indicator states.

Comparison: Proposed Detector vs. Commercial Detector

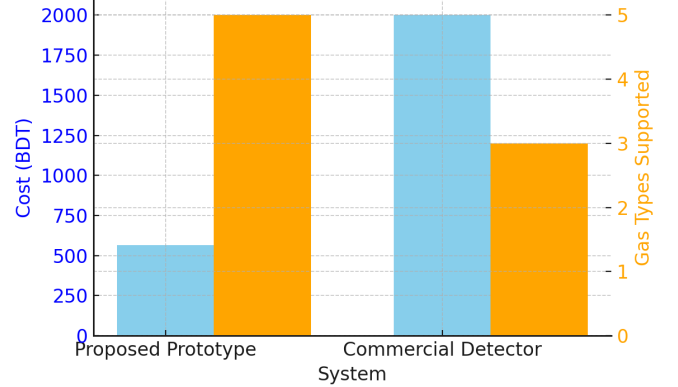


Fig. 4. Comparison of the proposed prototype and a commercial detector in terms of cost and supported gas types.

B. Cost and Feature Comparison

The combined cost of BOM was about 564 BDT, significantly lower than typical commercial detectors (often 2000 BDT). Furthermore, the commercial devices are unlike those of the commercial devices. the MQ-2 sensor accommodates a single gas (usually) optimized.

Fig. 4 compares cost and gas coverage between the proposed and commercial solutions. Among the proposed and commercial solutions, analyses. ions.

C. Limitations and Risk Controls

MOS sensors are affected by humidity, temperature, and cross-sensitivity. Calibration, periodic functional checks, and protective enclosures are essential for robust deployment. Incorporating hysteresis into comparators may further reduce false triggering.

VII. PRACTICAL IMPLICATIONS

In addition to academic prototyping, the suggested detector addresses urgent safety requirements in cost-sensitive contexts. It can avert disastrous fires in residential kitchens by early warning. It gives surveillance of in laboratories. volatile solvents. In automotive applications, detecting occurs. fuel vapor in confined compartments. Light industrial facilities is able to place numerous low-cost nodes to cover areas. The transparent design is also an educational platform of. teaching embedded sensing and instrumentation..

VIII. CONCLUSION AND FUTURE WORK

This contained a low cost multi-threshold gas leakage. MQ-2 sensor based and LM339 comparator detector. The prototype verified the discrete functional behavior. graphical and sound signals and remain at a low cost. Calibration against standardized will be done in the future. gas mixtures, remote alerts connected through IoT, rugged enclosure design, and compliance with certification standards such as IEC 60079-29. These guidelines will further the. system from proof-of-concept into a deployable safety device for residential, laboratory, and industrial domains.

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